

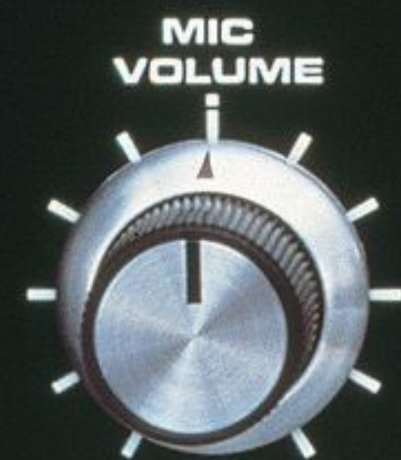
AUDIO / VIDEO TECH VL03



SPACE ECHO RE-201



PEAK
LEVEL



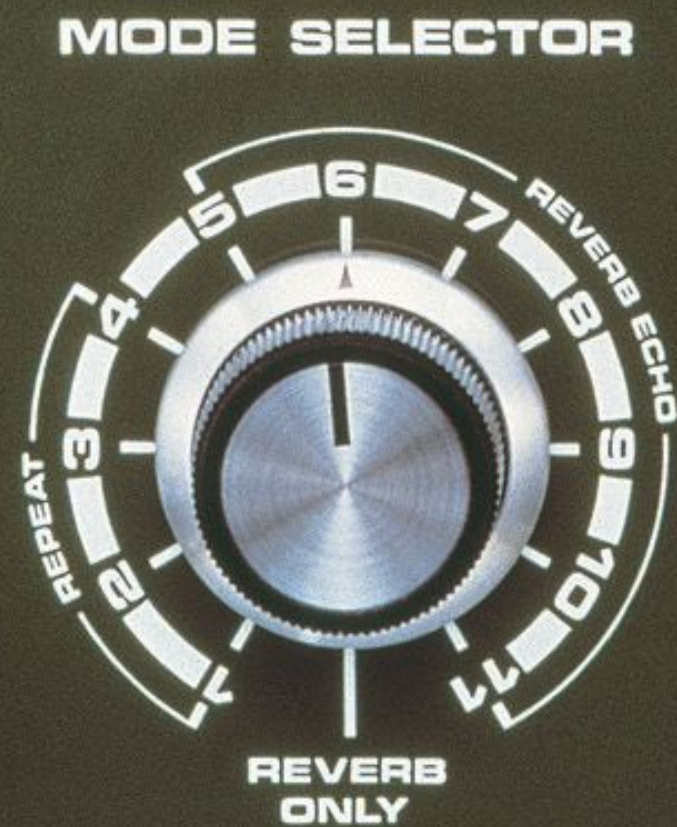
MIC
VOLUME



MIC
VOLUME



INSTRUMENT
VOLUME



MODE SELECTOR

REVERB
ONLY



BASS



TREBLE



REVERB VOLUME



REPEAT RATE



INTENSITY



ECHO VOLUME

STRAIGHT ECHO



ON
OFF



POWER

MIC



MIC



INSTRUMENT



ECHO NORMAL

FROM P.A.



OUT PUT



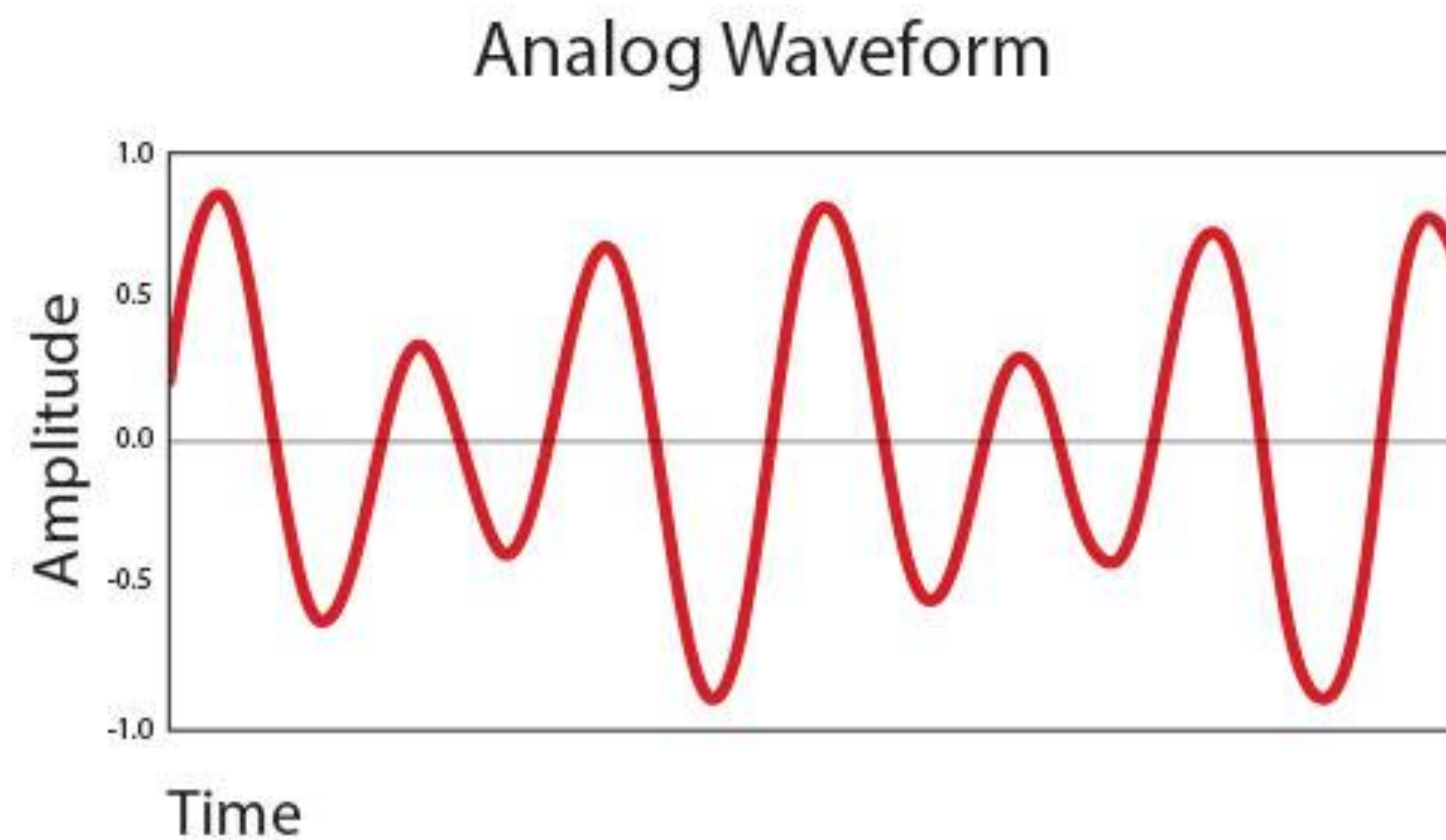
FOOT SW



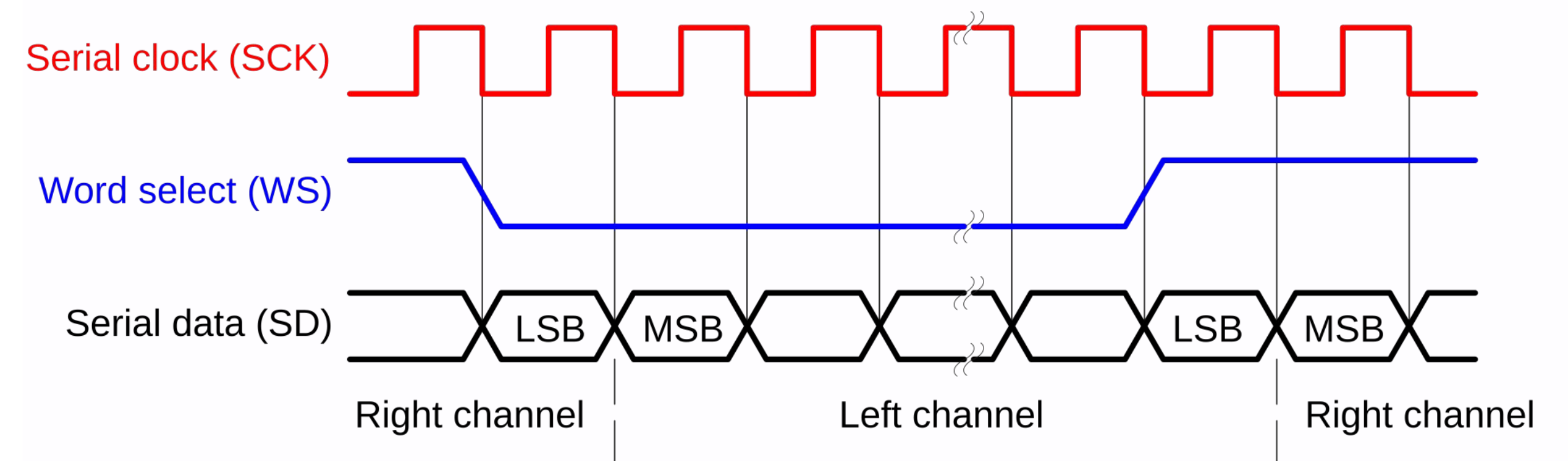
ECHO CANCEL

WIEDERHOLUNG

analog (zB. Spannung über Zeit,
geschlossener Stromkreis)



digital (zB. I2S,
serielle Übertragung)



ÜBERTRAGUNGSFORMATE
analog vs digital

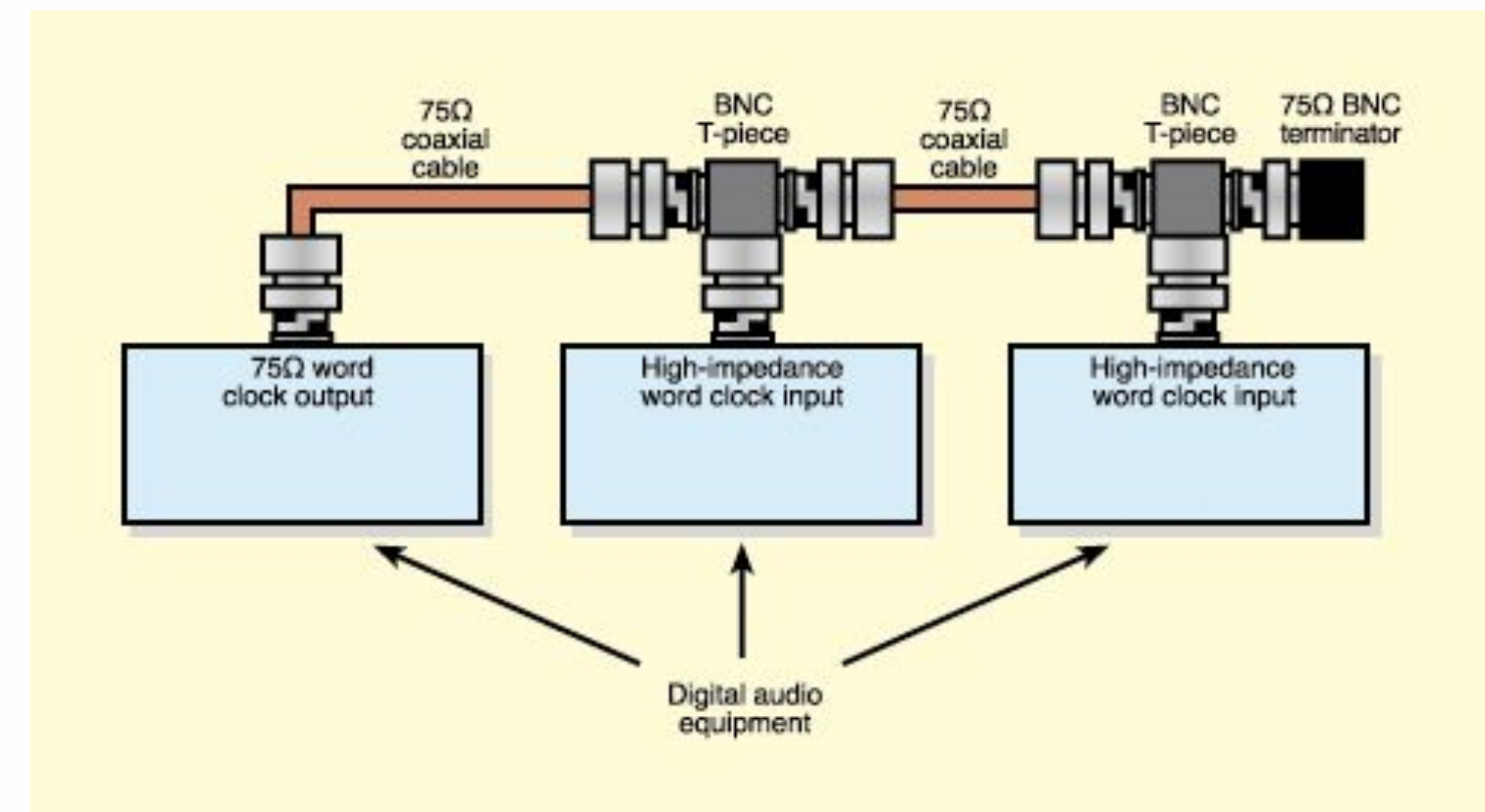
synchronous: sender and receiver use an extra connection for synchronisation
AoE, ADAT, AES

asynchronous: no dedicated synch signal, start and stop bits tell the receiver when a dataset starts and ends
MIDI, DMX

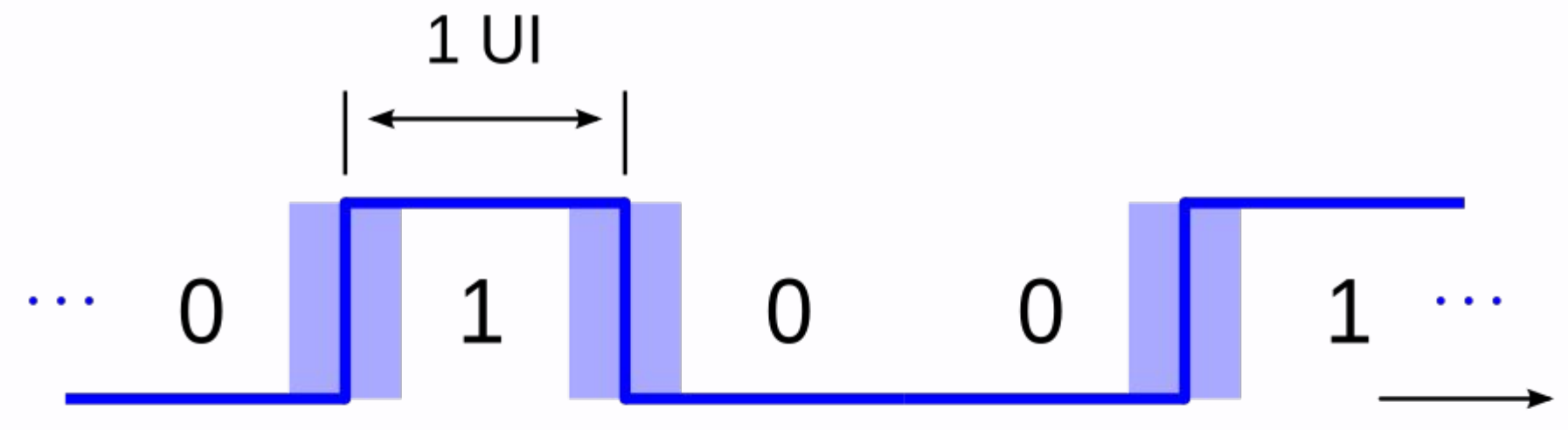
SYNCHRON VS ASYNCHRON

wordclock

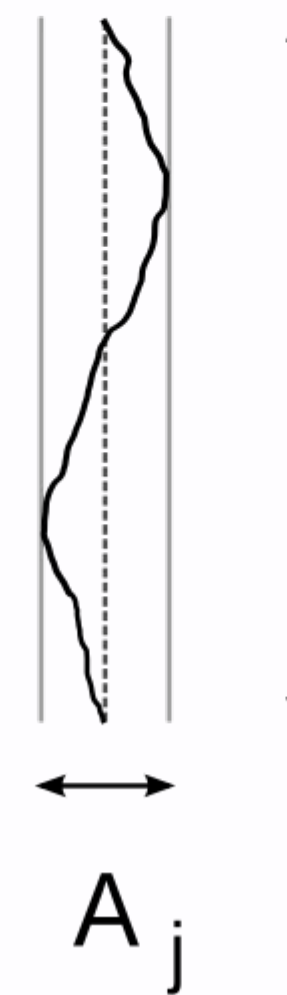
- signal to synchronize sender and receiver of a digital audio signal
- generated by one device (clock source) and distributed to one or more devices
- transmitted through dedicated word clock cables or within digital formats (AES, ADAT, AVB, etc.)
- 75Ohm impedance coax cables (BNC), must be terminated with a 75Ohm resistor to prevent reflecting the signal



ÜBERTRAGUNGSFORMATE
wordclock



- undesired deviation from a periodic signal (blue area vs blue line)
- comes from electromagnetic interference, crosstalk, DA or AD conversion
- can cause drop outs, distortion, clicks, (audible noise according to HIFI people?)
- low frequency phase shifts can be compensated for by phase-locked-loops (PPL)



UI: unit interval

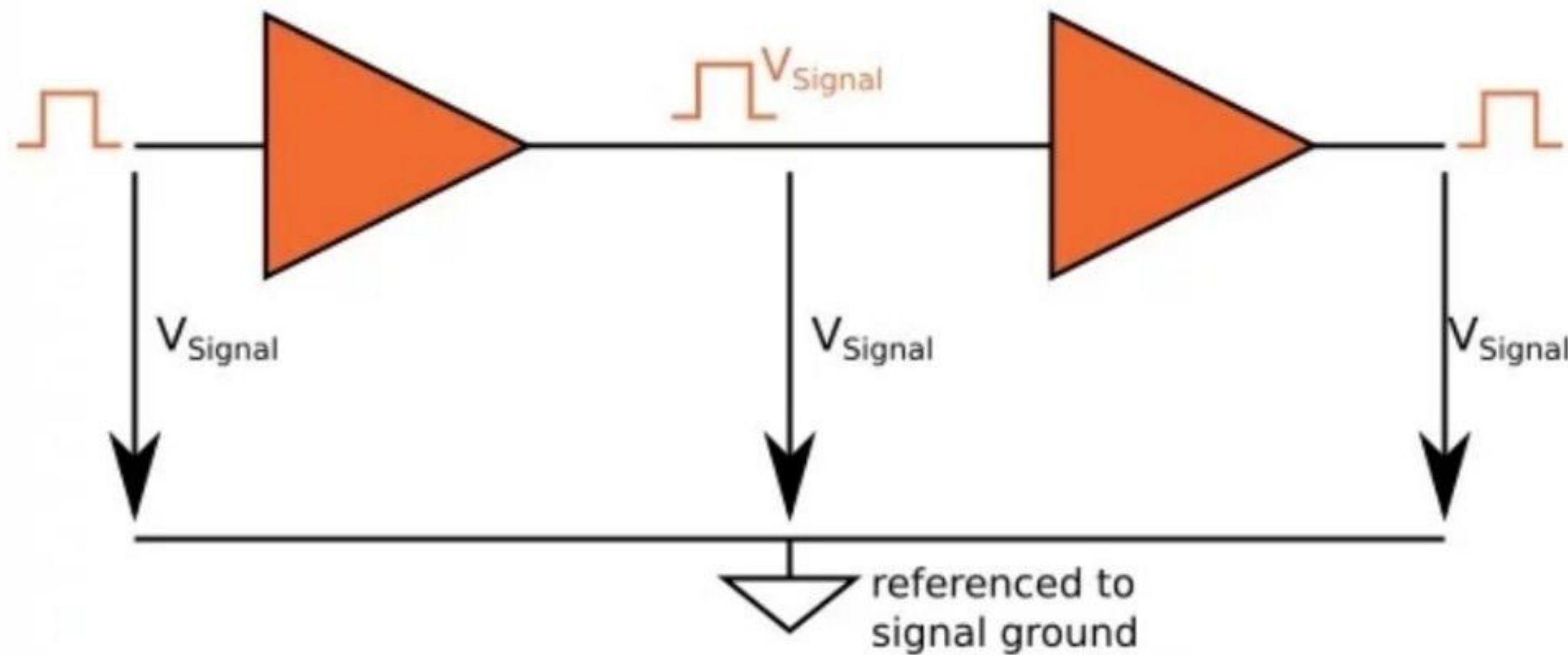
t: time

T_j : jitter period of base frequency

A_j : absolute jitter amplitude

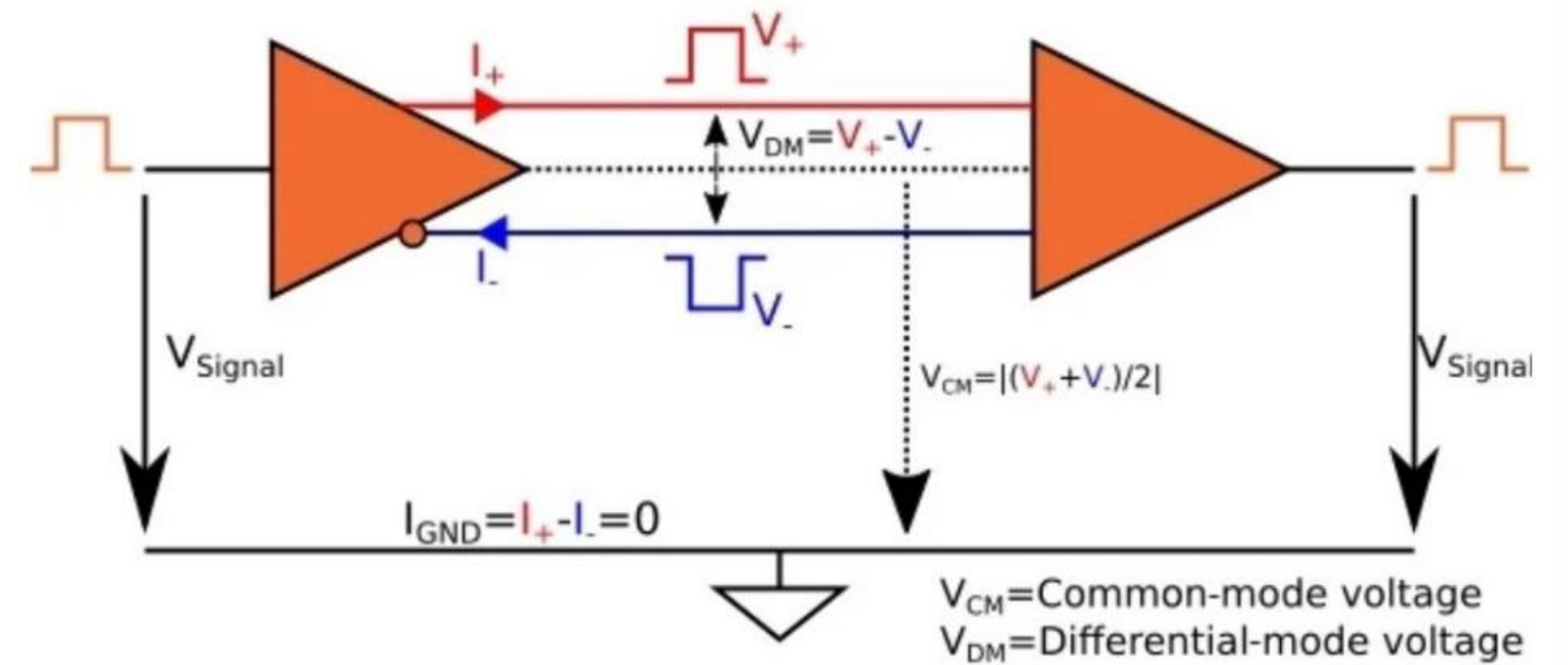
ÜBERTRAGUNGSFORMATE
jitter

Single-ended



unbalanced
(unsymmetrisch)

Differential



balanced
(symmetrisch)

ÜBERTRAGUNGSFORMATE
single ended vs differential



ÜBERTRAGUNGSFORMATE

Teil 1: analog

	XLR	KLINKE	CINCH	SPEAKON
NORM	3 POL	2/3 POL	2 POL (x2)	2/4/8 POL
FORMAT	MONO ANALOG SYMMETRISCH	ANALOG MONO SYM. STEREO UNSYM	ANALOG STEREO UNSYM	ANALOG BIS 4 KANAL UNSYMMETRISCH
TYP. ANWENDUNG	MIKROFON LINE	INSTRUMENT LINE	CONSUMER	LEISTUNG LAUTSPRECHER
ALT. ANWENDUNG	DIGITAL AES-3 LAUTSPRECHER	INSERTKABEL LAUTSPRECHER	DIGITAL S/PDIF	POWERCON
BEISPIELS PEGEL	MIKROFON $94\text{dB}_{\text{SPL}} = 1,85\text{mV}$	E-GITARRE Anschlag = 200mV	CONSUMER $-10\text{dB}_v = 0,3\text{V}$	$200\text{W Amp into } 4\Omega$ Speaker $\approx 7\text{V}$

ÜBERSICHT STECKER
ANALOG



ÜBERTRAGUNGSFORMATE
Teil 2: digital



STECKER DIGITAL

SPDIF

Sony/Philipps Digital InterFace

Kanalzahl 2 / Stereo

Bittiefe 16/20/24 bit

Samplerate 44,1/48kHz (max. 96kHz)

Kabel Koax (unsymmetrisch), optisch

Stecker Cinch, Toslink



DIGITAL SPDIF

AES/EBU | AES3

Audio Engineering Society /
European Broadcast Union

Kanalzahl 2 / Stereo

Bittiefe 16/20/24 bit

Samplerate max. 192kHz

Kabel 110 Ω twisted-pair (bal.), 75 Ω coax (unbal.)

Stecker XLR, BNC



DIGITAL AES3



DIGITALFORMATE

ADAT

Alesis Digital Audio Tape



Kanalzahl 8
Bittiefe max. 24 bit
Samplerate max. 48kHz (SMUX 4ch/96kHz, 2ch/192kHz)
Kabel optisch
Stecker Toslink

DIGITAL ADAT

MADI

Multi Channel Audio Digital Interface

Kanalzahl 64

Bittiefe 24 bit

Samplerate max. 48kHz (SMUX 32ch/96kHz, 16ch/192kHz)

Kabel koaxial, optisch

Stecker BNC, FDDI



DIGITAL MADI

	BNC	XLR	TOSLINK	FDDI
NORM	2 POL	3 POL	OPTISCH	OPTISCH
FORMAT	DIGITAL UNSYMMETRISCH	DIGITAL SYMMETRISCH	DIGITAL	DIGITAL
TYP. ANWENDUNG	WORDCLOCK	AES-3	S/PDIF	MADI
ALT. ANWENDUNG	AES-3 / MADI	S/PDIF (Analog MIC/LINE)	ADAT	NETZWERK TECHNIK

ÜBERSICHT STECKER
DIGITAL



DIGITALFORMATE

Audio over Ethernet

CobraNet, EtherSound, Dante, AVB, A-net, SoundGrid, AES50 etc.

channels: max 64x64 streams

bitrate: max. 32 bit

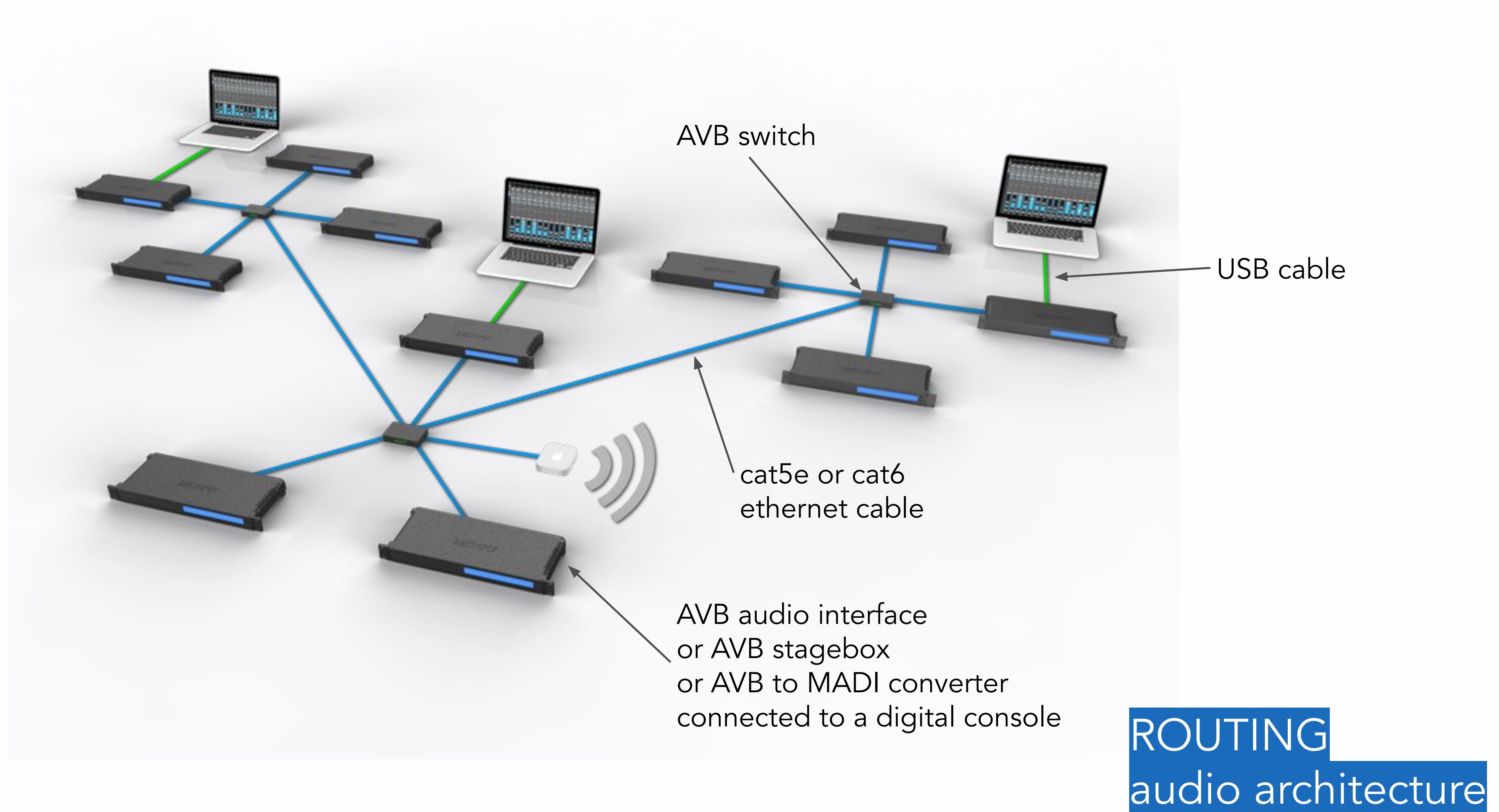
samplerate: max. 192kHz

cable: twisted-pair Cat5e, Cat6

plug: RJ45



AUDIONETWORK



[illegible]

◀

MOTU

Device

Routing

Mixing

Aux Mixing

Touch Console

DEVICES

UltraLite AVB

ENABLE INPUTS

Mic

Guitar

Analog

Optical

From Computer

16

ENABLE OUTPUTS

Phones

Main

Analog

Optical

To Mixer

To Computer

48

16

AVB CONFIGURATION

Input Streams

Output Streams

2

2

INPUT CONNECTIONS

1: None

2: None

ENABLE INPUTS

Mic

Guitar

Analog

Optical

From Computer

16

ENABLE OUTPUTS

Phones

Main

Analog

Optical

To Mixer

To Computer

48

16

AVB CONFIGURATION

Input Streams

Output Streams

2

2

INPUT CONNECTIONS

1: None

2: None

Output

Source

Phones L

Phones R

Main L

Main R

Analog 1

Analog 2

Analog 3

Analog 4

Analog 5

Analog 6

To Computer 1

To Computer 2

To Computer 3

To Computer 4

To Computer 5

To Computer 6

To Computer 7

To Computer 8

To Computer 9

To Computer 10

To Computer 11

To Computer 12

To Computer 13

To Computer 14

To Computer 15

To Computer 16

AVB Stream 1

AVB Stream 2

In 1

In 2

In 3

In 4

In 5

In 6

In 7

In 8

In 9

In 10

In 11

In 12

In 13

In 14

In 15

Source

Monitor L

Monitor R

Main L

Main R

Aux 1

Aux 2

Aux 3

Aux 4

Aux 5

Aux 6

Mic 1

Mic 2

Guitar 1

Guitar 2

Analog 1

Analog 2

Analog 3

Analog 4

Analog 5

Analog 6

From Computer 1

From Computer 2

From Computer 3

From Computer 4

From Computer 5

From Computer 6

From Computer 7

From Computer 8

From Computer 9

From Computer 10

From Computer 11

From Computer 12

From Computer 13

From Computer 14

From Computer 15

From Computer 16

AVB Stream 1

AVB Stream 2

Main L

Main R

Monitor L

Monitor R

Aux 1

Aux 2

Aux 3

Aux 4

Aux 5

Aux 6

Aux 7

Aux 8

Aux 9

Aux 10

Aux 11

Aux 12

Aux 13

Aux 14

Mix Group

Mix Reverb

Inputs

Analog

From Computer (USB)

Mix Aux

ROUTING
local IOs + mixer

MOTU

Device

Routing

Mixing

Aux Mixing

Touch Console

DEVICES

UltraLite AVB

ENABLE INPUTS

Mic

Guitar

Analog

Optical

From Computer

16

ENABLE OUTPUTS

Phones

Main

Analog

Optical

To Mixer

10

To Computer

16

AVB CONFIGURATION

Input Streams

2

Output Streams

2

INPUT CONNECTIONS

1: None

2: None

ENABLE INPUTS

Mic

Guitar

Analog

Optical

From Computer

16

ENABLE OUTPUTS

Phones

Main

Analog

Optical

To Mixer

10

To Computer

16

AVB CONFIGURATION

Input Streams

2

Output Streams

2

INPUT CONNECTIONS

1: None

2: None

Output

Source

Phones L

Phones R

Main L

Main R

Analog 1

Analog 2

Analog 3

Analog 4

Analog 5

Analog 6

To Computer 1

To Computer 2

To Computer 3

To Computer 4

To Computer 5

To Computer 6

To Computer 7

To Computer 8

To Computer 9

To Computer 10

To Computer 11

To Computer 12

To Computer 13

To Computer 14

To Computer 15

To Computer 16

AVB Stream 1 1

AVB Stream 1 2

AVB Stream 1 3

AVB Stream 1 4

AVB Stream 1 5

AVB Stream 1 6

AVB Stream 1 7

AVB Stream 1 8

AVB Stream 2 1

AVB Stream 2 2

AVB Stream 2 3

AVB Stream 2 4

AVB Stream 2 5

AVB Stream 2 6

AVB Stream 2 7

AVB Stream 2 8

Mix In

Mic 1

Mic 2

Guitar 1

Guitar 2

Analog 1

Analog 2

Analog 3

Analog 4

Analog 5

Analog 6

From Computer 1

From Computer 2

From Computer 3

From Computer 4

From Computer 5

From Computer 6

From Computer 7

From Computer 8

From Computer 9

From Computer 10

From Computer 11

From Computer 12

From Computer 13

From Computer 14

From Computer 15

From Computer 16

AVB Stream 1 1

AVB Stream 1 2

AVB Stream 1 3

AVB Stream 1 4

AVB Stream 1 5

AVB Stream 1 6

AVB Stream 1 7

AVB Stream 1 8

AVB Stream 2 1

AVB Stream 2 2

AVB Stream 2 3

AVB Stream 2 4

AVB Stream 2 5

AVB Stream 2 6

AVB Stream 2 7

AVB Stream 2 8

Mix Main

Mix Monitor

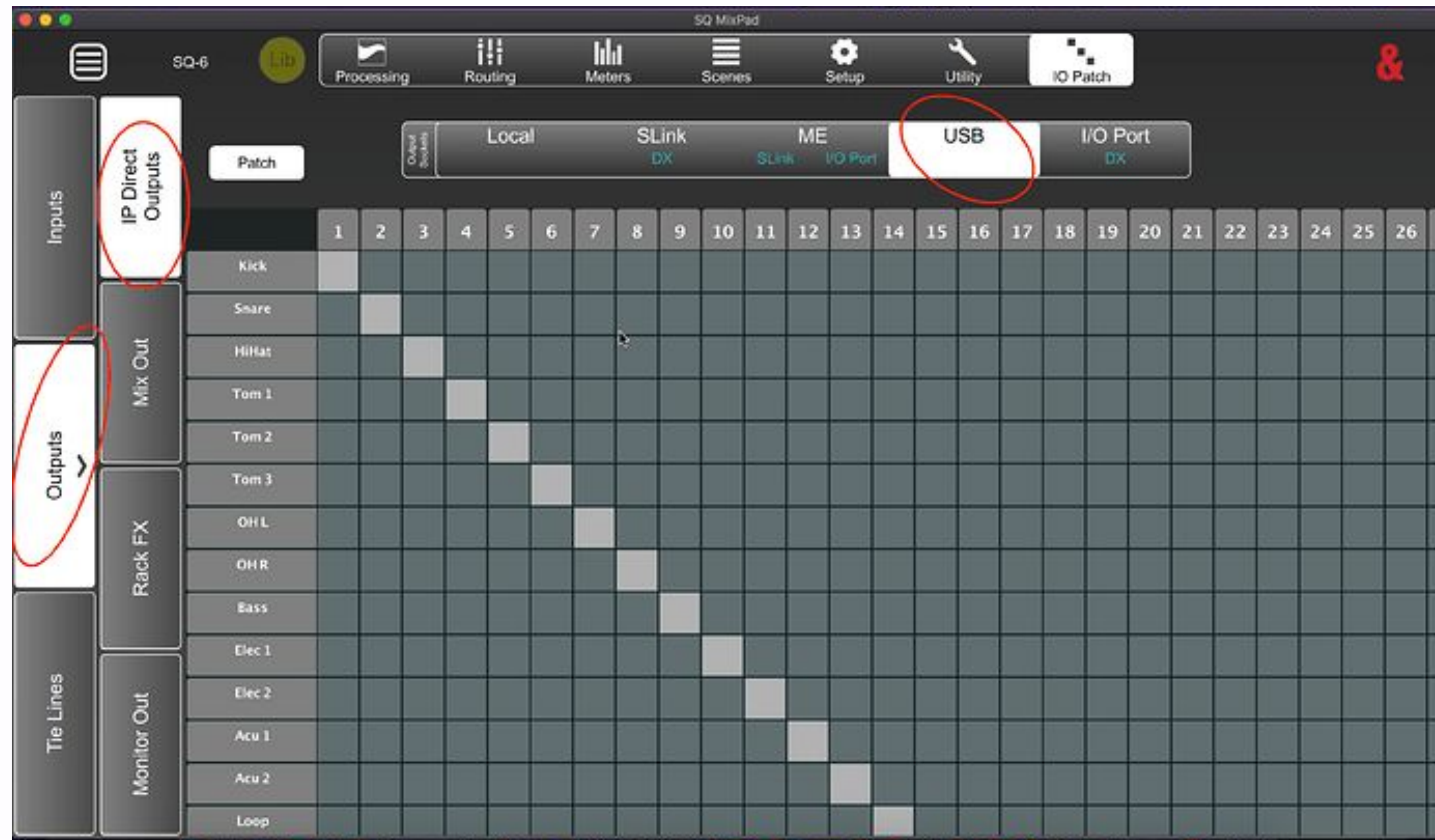
Mix Aux

Mix Group

Mix Reverb

Mix Post FX

ROUTING
AVB stream



ROUTING
A&H SQ series

